REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

The Many Forms of Energy Practice Questions

Grade 12 Advanced | Module 10 Lesson 2 | The Many Forms of Energy

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: $KE = \frac{1}{2}mv^2$; $KE_{rot} = \frac{1}{2}I\omega^2$; $GPE = mgh$; $PE_{spring} = \frac{1}{2}kx^2$; $E_{rest} = mc^2$; $g = 9.8\ m/s^2$.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 14–22.

P01. Translational kinetic energy is proportional to:

A. Speed only B. The inverse of mass
C. The square of speed D. The square of time in every situation

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P02. Gravitational potential energy depends on the chosen:

A. Coordinate unit only with no conversion B. Object colour
C. Direction of horizontal travel only D. Zero-height reference

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P03. Elastic potential energy can be stored in a:

A. Stretched spring B. Stationary uncompressed spring at its defined minimum only
C. Reference point with no object D. Unit conversion

P04. Rest energy is related to mass by:

A. $E = m/c^2$ B. $E = mc^2$
C. $E = mv$ D. $E = mg$ only

P05. For the same I, doubling angular speed makes rotational kinetic energy:

A. Twice as large B. Half as large
C. Four times as large D. Unchanged

P06. A 4 kg object moves at 3.0 m/s. Find its translational kinetic energy.

A. 18 J B. 36 J
C. 9 J D. -18 J

P07. A 4 kg object is 3.0 m above the chosen zero level. Find its gravitational potential energy. Use $g = 9.8\ m/s^2$.

A. 235.2 J B. 117.6 J
C. 58.8 J D. -117.6 J

P08. A rotor has moment of inertia 0.4 kg m² and angular speed 5.0 rad/s. Find its rotational kinetic energy.

A. 10 J B. 2.5 J
C. 5 J D. -5 J

P09. A spring with constant 200 N/m is stretched 0.100 m from equilibrium. Find its elastic potential energy using PE = ½kx².

A. 2 J B. 0.5 J
C. -1 J D. 1 J

P10. Find the rest energy of mass 2×10^{-9} kg. Use $E = mc^2$ and $c = 3.00 \times 10^8$ m/s.

A. 1.8×10^8 J B. 3.6×10^8 J
C. 9×10^7 J D. -1.8×10^8 J

P11. Define potential energy.

P12. Define gravitational potential energy near Earth's surface.

P13. What is elastic potential energy?

P14. Explain the meaning of rest energy.

P15. Relate work done by gravity to the change in gravitational potential energy.

P16. A 6 kg object moves at 3.0 m/s. Find its translational kinetic energy.

P17. A 8 kg object moves at 3.0 m/s. Find its translational kinetic energy.

P18. A 6 kg object is 3.0 m above the chosen zero level. Find its gravitational potential energy. Use g = 9.8 m/s².

P19. A 8 kg object is 3.0 m above the chosen zero level. Find its gravitational potential energy. Use g = 9.8 m/s².

P20. A rotor has moment of inertia 0.6 kg m² and angular speed 5.0 rad/s. Find its rotational kinetic energy.

P21. A rotor has moment of inertia 0.8 kg m² and angular speed 5.0 rad/s. Find its rotational kinetic energy.

P22. A spring with constant 300 N/m is stretched 0.100 m from equilibrium. Find its elastic potential energy using PE = ½kx².

P23. A spring with constant 400 N/m is stretched 0.100 m from equilibrium. Find its elastic potential energy using PE = ½kx².

P24. Find the rest energy of mass 3×10^{-9} kg. Use $E = mc^2$ and $c = 3.00 \times 10^8$ m/s.

P25. Find the rest energy of mass 4×10^{-9} kg. Use $E = mc^2$ and $c = 3.00 \times 10^8$ m/s.

P26. A 2.0 kg object moves horizontally at 4.0 m/s at a height 3.0 m above the reference. Find KE and GPE.

P27. An object moves from height 5.0 m to 2.0 m. Explain the signs of ΔGPE and gravitational work.

P28. If the GPE zero level is moved upward, does the physical gravitational work between two fixed heights change?

P29. Compare the energy changes when a spring's extension doubles and when an object's speed doubles.

P30. A wheel rolls without slipping. Why can its kinetic energy have two terms?